AI-Powered Smart Farming Robot

Disease Detection & Automated Fertilizer Spraying
First Progress Presentation

HND SOFTWARE ENGINEERING 2026

PROJECT TEAM

AI-Powered Smart Farming Robot for Disease Detection and Automated Fertilizer Spraying

Programme

Higher National Diploma in Software Engineering

Presentation Type

First Progress Presentation

Supervisor

(Lecturer Name)

Project Goal

Develop an intelligent farming robot that identifies plant diseases using AI and automatically applies suitable treatments — improving agricultural productivity and reducing manual effort.

BACKGROUND

Introduction

Agriculture is a cornerstone of Sri Lanka's economy, yet farmers face persistent challenges: plant diseases, labour shortages, improper fertiliser usage, delayed identification, and reduced crop yields.

Traditional disease detection relies on human observation often inaccurate and time-consuming. Advancements in AI, robotics, and IoT now create opportunities to automate farming activities.

This project proposes an intelligent farming robot capable of detecting diseases, classifying disease types, and automatically spraying appropriate treatments making farming more efficient, accurate, and sustainable.

Problem Statement

Manual Detection

Farmers identify diseases by eye — slow, inconsistent, and prone to misdiagnosis.

Late Identification

Symptoms often go unnoticed until damage is irreversible, allowing rapid spread.

Chemical Misuse

Incorrect pesticide application damages crops and harms the environment through overuse.

Labour & Cost

Manual spraying demands significant labour; large farms are difficult to monitor continuously.

An automated system is urgently needed to detect diseases early and deliver accurate, targeted treatment.

GOALS

Project Aim & Objectives

Main Aim: Design and develop an AI-powered smart farming robot capable of detecting crop diseases and automatically applying suitable treatments.

Platform & Data

- Develop a mobile robotic platform
- Collect and prepare crop disease datasets
- Train an AI model for disease classification
- Differentiate bacterial and fungal diseases

Intelligence & Action

- Capture crop images using a camera module
- Analyse leaf images using machine learning
- Generate treatment recommendations
- Automate fertiliser spraying
- Reduce labour and improve farming efficiency

DATASET

Dataset Selection

A dataset was selected based on commonly cultivated vegetable crops in Sri Lanka, covering both highland and lowland varieties to ensure regional relevance and data diversity.

Highland Vegetables

- Carrots
- Leeks
- Cabbage
- Potatoes
- Beans

Lowland Vegetables

- Brinjal
- Pumpkin
- Okra
- Tomatoes
- Green Chilies
- Red Onions
- Gourds

These crops were chosen for their economic importance, widespread cultivation, and susceptibility to common bacterial and fungal diseases providing sufficient diversity for AI training.

Disease Categories in the Dataset

Healthy Plants

Normal crop conditions with no disease symptoms — serving as the baseline for classification.

Bacterial Diseases

Bacterial Wilt, Bacterial Spot, Soft Rot. Symptoms include leaf yellowing, water-soaked lesions, and wilting.

Fungal Diseases

Early Blight, Powdery Mildew, Leaf Mold. Symptoms include brown spots, mould growth, and white powdery deposits.

The AI model will learn to distinguish these three categories automatically from the labelled training images.

APPROACH

Why Bacterial & Fungal Classification?

Many existing disease detection systems identify only specific diseases. Our approach focuses on broader classification distinguishing bacterial from fungal infections enabling faster, more scalable automation.

Fast Decisions

Simpler Training

Targeted Treatment

Scalable

Practical Advantages

- Faster decision-making in the field
- Easier model training with clearer categories
- Simplified treatment selection logic
- Better scalability across crop types
- Reduced computational complexity

Once a disease is classified as bacterial or fungal, the robot automatically selects the appropriate treatment improving automation and reducing dependence on expert intervention.

SYSTEM DESIGN

Proposed System Overview

Image Acquisition

Camera module captures leaf images during patrol.

AI Processing

CNN model detects and classifies disease categories.

Decision Engine

Generates treatment recommendation based on classification.

Robotic Platform

Navigates farming areas autonomously between crop rows.

Spraying Unit

Applies the selected treatment precisely on affected plants.

User Interface

Enables monitoring, control, and real-time feedback for farmers.

ARCHITECTURE

System Architecture

- Capture
- AI Ingest
- Preprocess
- CNN Analyze

The architecture follows a linear pipeline from image capture to treatment delivery, with data logging at each stage for continuous improvement and farmer visibility.

Thank You!

We appreciate your time and attention. We are happy to answer any questions.